MATH 463 - INTEGRAL EQUATIONS

LESSON 5

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OUTLINE

1. GREEN'S FUNCTION
   - Converting a Boundary Value Problem to a Fredholm's Integral Equation
   - Green's Function
   - Exercises

CONVERTING A BOUNDARY VALUE PROBLEM TO A FREDHOLM'S INTEGRAL EQUATION

TYPE 1(No derivative in the boundary condition)

Let $y''(x) + g(x)y(x) = h(x)$ be a boundary value problem with conditions $y(0) = \alpha, y(1) = \beta$ for $0 < x < 1$

Let $y'' = u(x)$ (1)

$$\int_0^x y''(t)\ dt = \int_0^x u(t)\ dt$$

$$y'(x) - y'(0) = \int_0^x u(t)\ dt$$

$$\int_0^x y'(t)\ dt - \int_0^x y'(0)\ dt = \int_0^x \int_0^x u(t)\ dtdt$$

$$y(x) = y(0) + xy'(0) + \int_0^x \int_0^x u(t)\ dtdt$$

$$y(x) = \alpha + xy'(0) + \int_0^x (x-t)u(t)\ dt$$

When $x = 1$

$$y(1) = \alpha + y'(0) + \int_0^1 (1-t)u(t)\ dt$$

$$\beta = \alpha + y'(0) + \int_0^1 (1-t)u(t)\ dt$$

$$y'(0) = \beta - \alpha - \int_0^1 (1-t)u(t)\ dt$$

$$\implies y(x) = \alpha + x\left[\beta - \alpha - \int_0^1 (1-t)u(t)\ dt\right] + \int_0^x (x-t)u(t)\ dt$$

$$y(x) = \alpha + (\beta - \alpha)x + \int_0^1 (xt - x)u(t)\ dt + \int_0^x (x-t)u(t)\ dt$$

$$y(x) = \alpha + (\beta - \alpha)x + \int_0^x (xt - x)u(t)\ dt + \int_x^1 (xt - x)u(t)\ dt + \int_0^x (x-t)u(t)\ dt$$

$$y(x) = \alpha + (\beta - \alpha)x + \int_0^x (xt - t)u(t)\ dt + \int_x^1 (xt - x)u(t)\ dt \quad (2)$$

put (1) and (2) into the boundary value problem

$$u(x)+g(x)\left[\alpha+(\beta-\alpha)x+\int_0^x (xt-t)u(t)\ dt+\int_x^1 (xt-x)u(t)\ dt\right]=h(x)$$

$$u(x)=h(x)-\alpha g(x)+(\alpha-\beta)xg(x)+g(x)\int_0^x (t-xt)u(t)\ dt+$$

$$g(x)\int_x^1 (x-xt)u(t)\ dt$$

$$u(x)=h(x)-\alpha g(x)+(\alpha-\beta)xg(x)+\int_0^x t(1-x)g(x)u(t)\ dt+$$

$$\int_x^1 x(1-t)g(x)u(t)\ dt$$

which is of the form

$$u(x)=f(x)+\int_0^1 K(x,t)u(t)\ dt$$

where

$$f(x) = h(x) - \alpha g(x) + (\alpha - \beta)xg(x)$$

$$K(x,t) = \begin{cases} t(1-x)g(x) & 0 \leq t \leq x \\ x(1-t)g(x) & x \leq t \leq 1 \end{cases}$$

NOTE:

1. If the boundary value problem is homogeneous, the resulting Fredholm's integral equation will also be homogeneous.
2. The resulting kernel should be a continuous function.

Example:

Convert the boundary value problem into a Fredholm's integral equation.

$$y''(x) + 4y(x) = \sin x \qquad y(0) = y(1) = 0 \qquad \textit{for } 0 < x < 1$$

Solution:

$$y''(x) = u(x) \tag{1}$$

$$\int_0^x y''(t)\ dt = \int_0^x u(t)\ dt$$

$$y'(x) = y'(0) + \int_0^x u(t)\ dt$$

$$\int_0^x y'(t)\ dt = \int_0^x y'(0)\ dt + \int_0^x \int_0^x u(t)\ dt$$

$$y(x) = y(0) + xy'(0)t + \int_0^x \int_0^x u(t)\ dt$$

$$y(x) = xy'(0) + \int_0^x (x-t)u(t)\ dt$$

when $x = 1$

$$y(1) = y'(0) + \int_0^1 (1-t)u(t)\ dt$$

$$y'(0) = -\int_0^1 (1-t)u(t)\ dt$$

$$\implies y(x) = \int_0^1 x(t-1)u(t)\ dt + \int_0^x (x-t)u(t)\ dt$$

$$y(x) = \int_0^x x(t-1)u(t)\ dt + \int_x^1 x(t-1)u(t)\ dt + \int_0^x (x-t)u(t)\ dt$$

$$y(x) = \int_0^x t(x-1)u(t)\ dt + \int_x^1 x(t-1)u(t)\ dt \qquad (2)$$

put (1) and (2) into the boundary value problem

$$u(x) + 4\left[\int_0^x t(x-1)u(t)\ dt + \int_x^1 x(t-1)u(t)\ dt\right] = \sin x$$

$$u(x) = \sin x + \int_0^x 4t(1-x)u(t)\ dt + \int_x^1 4x(1-t)u(t)\ dt$$

$$u(x) = f(x) + 4\int^1 K(x,t)u(t)\ dt$$

where

$$f(x) = \sin x$$

$$K(x,t) = \begin{cases} t(1-x) & 0 \leq t \leq x \\ x(1-t) & x \leq t \leq 1 \end{cases}$$

testing for the continuity of the kernel

$$K_1(x,t) = t(1-x) \ and \ K_2(x,t) = x(1-t)$$

replace x with t in both kernels

$$K_1(t,t) = t(1-t) \ and \ K_2(t,t) = t(1-t)$$

Since both kernels are the same, it is continuous.

GREEN'S FUNCTION

Given $L(y) = 0$, a homogeneous linear differential equation with boundary conditions $a < x < b$, $y(a) = \alpha$, $y(b) = \beta$ where L is a differential operator and is given as

$$L = p_0(x)\frac{d^n}{dx^n} + p_1(x)\frac{d^{n-1}}{dx^{n-1}} + ... + p_n(x)$$

.

Example:

$$L(y) = y'' + 4y(x)$$

In this case,

$$L = \frac{d^2}{dx^2} + 4$$

If $y(x) = 0$ is the trivial solution to the boundary value problem, then there exist a Green's function $G(x, t)$ for the boundary value problem.

For $G(x,t)$ to qualify as Green's function, it must satisfy the following four(4) conditions.

1. $G(x,t)$ satisfies $L(G) = 0$
2. $G(x,t)$ satisfies the conditions at the boundaries i.e $G(x,t)\Big|_{x=a,x=b} = \alpha, \beta$. Taking $G(x,t)$ as a function of x.
3. $G(x,t)$ must be continuous within the boundaries including all its derivatives up to $(n-2)th$ order.
4. The $(n-1)th$ order derivative of $G(x,t)$ has a discontinuity of the first kind (jump discontinuity) i.e $R.H\ limit \neq L.H\ limit$ with magnitude of $-\dfrac{1}{p_0(t)}$. Where p_0 is the coefficient of the highest derivative of $y(x)$ i.e $\dfrac{\delta^{n-1}G(x,t)}{\delta x^{n-1}}\Big|_{x=t} = -\dfrac{1}{p_0(t)}$

Generally $L(y) + \phi(x) = 0$ with some boundary conditions.

1. If ϕ is a function of only x i.e $\phi = \phi(x)$, then the solution of the boundary value problem is

$$y(x) = \int_a^b G(x,t)\phi(t)\ dt$$

2. If ϕ is a function of both x and y i.e $\phi = \phi(x, y(x))$ then

$$y(x) = \int_a^b G(x,t)\phi(x, y(x))\ dt$$

is the corresponding Fredholm integral equation of the boundary value problem.

Special Case (2nd Order Differential Equation)

This is how the four(4) conditions will look like:

1. $L(G) = 0$ where

$$L = p_0(x)\frac{d^2}{dx^2} + p_1(x)\frac{d}{dx} + p_2(x)$$

2. $G_1(x,t)|_{x=a} = \alpha$
   $G_2(x,t)|_{x=b} = \beta$
3. $G_1(t,t) = G_2(t,t)$
4. $G_2'(t) - G_1'(t) = -\dfrac{1}{p(t)}$

Example

Construct the Green's function of the boundary value problem $y''(x) = 0$
$y(0) = y(l) = 0$

Solution:

$L = \dfrac{d^2}{dx^2}$ where $p = 1$

$$y''(x) = 0$$

$$\int y''(x)\ dx = \int 0\ dx$$

$$\int y'(x)\ dx = \int A\ dx$$

$$y(x) = Ax + B$$

at $x = 0$

$$B = 0$$

$$\implies y(x) = Ax$$

at $x = l$

$$y(l) = Al = 0$$

$$A = 0$$

Hence $y(x) = 0$ is a trivial solution to the boundary value problem. Hence there exist a Green's function $G(x, t)$
since $y(x) = Ax + b$, we choose our Green's function to be

$$G(x,t) = \begin{cases} ax+b & 0 \leq x < t \\ cx+d & t < x \leq l \end{cases}$$

It satisfies the first condition $L(G) = 0$
Testing for the second condition i.e $G(x,t)|_{x=0,x=l} = 0$

$$G_1(x,t) = ax + b$$

$$G_1(0,t) = a(0) + b = 0$$

$$b = 0$$

$$G_1(x,t) = ax \tag{1}$$

$$G_2(x,t) = cx + d$$

$$G_2(l,t) = cl + d \tag{2}$$

$$cl + d = 0 \tag{3}$$

Checking for the third(3) condition:

$$G_1(t,t) = G_2(t,t)$$

$$at = ct + d$$

$$(a-c)t = d \tag{4}$$

Checking for the fourth(4th) condition:

$$G_2'(t) - G_1'(t) = -\frac{1}{p(t)} \qquad at\ x = t$$

$$c - a = -1$$

$$a - c = 1 \tag{5}$$

put (5) into (4)

$$d = t \tag{6}$$

put (6) into (3)

$$c = -\frac{t}{l} \tag{7}$$

put (7) into (5)

$$a = \frac{l-t}{l}$$

$$G_1(x,t) = \frac{l-t}{l}x = \frac{x}{l}(l-t)$$

$$G_2(x,t) = -\frac{t}{l}x + t = \frac{t}{l}(l-x)$$

$$G(x,t) = \begin{cases} \frac{x}{l}(l-t) & 0 \leq x < t \\ \frac{t}{l}(l-x) & t < x \leq l \end{cases}$$

Example

Construct the Green's function for the following boundary value problem and use it to solve the boundary value problems. $y''(x) + y(x) = x$ where $y(0) = y(\frac{\pi}{2}) = 0$

Solution:

Expressing the problem in terms of

$$L(y) + \phi(x) = 0$$

where

$$L(y) = y''(x) + y(x) \text{ and } \phi(x) = -x$$

solving for y

$$y''(x) + y(x) = 0$$

$$r^2 + 1 = 0$$

$$r = \pm 1i$$

$$y(x) = Ae^{rx} = Ae^{\pm 1ix}$$

$$y(x) = A\cos(x) + B\sin(x)$$

at $x = 0$

$$y(0) = A = 0$$

Hence $y(x) = B\sin(x)$
at $x = \frac{\pi}{2}; B = 0$
Hence $y(x) = 0$.
Hence we can construct a Green's function for the boundary value problem.
Since $L(y) = 0$, we can choose a function similar to $y(x)$ as our Green's function such that $L(G) = 0$

$$G(x,t) = \begin{cases} A\cos(x)+B\sin(x) & 0 \leq x < t \\ C\cos(x)+D\sin(x) & t < x \leq \frac{\pi}{2} \end{cases}$$

$$G_1(x,t) = A\cos(x) + B\sin(x)$$

$$G_2(x,t) = C\cos(x) + D\sin(x)$$

from property 2:

$$G_1(0,t) = A\cos(0) + B\sin(0) = 0, \qquad A = 0$$

$$G_2\left(\frac{\pi}{2}, t\right) = C\cos\left(\frac{\pi}{2}\right) + D\sin\left(\frac{\pi}{2}\right) = 0, \qquad D = 0$$

Hence

$$G_1(x,t) = B\sin(x) \tag{1}$$

$$G_2(x,t) = C\cos(x) \tag{2}$$

from property 3:

$$G_1(t,t) = G_2(t,t)$$

$$B\sin(t) = C\cos(t)$$

$$B = \frac{C\cos(t)}{\sin(t)}$$

from property 4:

$$G_2'(x,t)|_{x=t} - G_1'(x,t)|_{x=t} = \frac{-1}{p_0(t)}$$

$$-C\sin(t) - B\cos(x) = -1$$

$$C\sin(t) + B\cos(x) = 1 \tag{4}$$

putting (3) into (4)

$$C\sin(t) + \frac{C\cos^2(t)}{C\sin(t)} = 1$$

$$C\sin^2(t) + C\cos^2(t) = \sin(t)$$

$$C = \sin(t)$$

putting the value of C into (3)

$$B = \cos(t)$$

putting the value of B and C into (1) and (2)

$$G_1(x,t) = \cos(t)\sin(x)$$

$$G_2(x,t) = \sin(t)\cos(x)$$

Hence the Green's function is

$$G(x,t) = \begin{cases} \cos(t)\sin(x) & 0 \leq x < t \\ \sin(t)\cos(x) & t < x \leq \frac{\pi}{2} \end{cases}$$

Since $\phi(x)$ is a function of only x, the solution to the boundary value problem is given by

$$y(x) = \int_0^{\frac{\pi}{2}} G(x,t)\phi(t)\ dt$$

$$y(x) = -\int_0^x G_1(x,t)t\ dt - \int_x^{\frac{\pi}{2}} G_2(x,t)t\ dt$$

NOTE:

When substituting the values of $G_1(x,t)$ and $G_2(x,t)$ into the equations, we exchange the positions of x and t such that $G_1(x,t)$ becomes $G_1(t,x)$ and $G_2(x,t)$ becomes $G_2(t,x)$

Hence

$$y(x) = -\int_0^x G_1(t,x)t\ dt - \int_x^{\frac{\pi}{2}} G_2(t,x)t\ dt$$

$$y(x) = -\int_0^x \cos(x)\sin(t)t\ dt - \int_x^{\frac{\pi}{2}} \sin(x)\cos(t)t\ dt$$

$$y(x) = -\cos(x)\int_0^x \sin(t)t\ dt - \sin(x)\int_x^{\frac{\pi}{2}} \cos(t)t\ dt$$

$$y(x) = -\cos(x)\left[-t\cos(t)\Big|_0^x + \sin(t)\Big|_0^x\right] -$$

$$\sin(x)\left[t\sin(t)\Big|_x^{\frac{\pi}{2}} + \cos(t)\Big|_x^{\frac{\pi}{2}}\right]$$

$$y(x) = -\cos(x)[-x\cos(x) + \sin(x)] - \sin(x)[\frac{\pi}{2} - x\sin(x) - \cos(x)]$$

$$y(x) = x - \frac{\pi}{2}\sin(x)$$

Exercises

$$y'' + 9y = \cos(x); \qquad y(0) = y(1) = 0; \qquad 0 < x < 1$$

$$y'' + 4y = \sin(x); \qquad y(0) = y(1) = 0; \qquad 0 < x < 1$$

Given the boundary value problems above;

1. Check if it is possible to construct a Green's function for each of them.
2. Construct the Green's function in the case where it is possible to do so.
3. Use the Green's function to compute the solution of the boundary value problem.